

SKILLS

Identity & Security: OAuth 2.0, OIDC, IAM, SSO, Zero-Trust, GDPR, HIPAA
Cloud & Developer Tooling: TypeScript, Python, Go, Node.js, AWS, GCP, Kubernetes, Terraform, Docker
Protocols & APIs: MCP, A2A, REST APIs, GraphQL, SDKs, sample apps
Data & Observability: OpenTelemetry, Kafka, ELK, Prometheus, tracing, alerting
Architecture & GenAI: LLMs, RAG, LangChain, embeddings, vector DBs, architecture
Delivery & Enablement: proofs of concept, SDKs, reference implementations, writing, workshops

EXPERIENCE

- Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect Nov 2022 – Present
 - ASB Builder Platform:** Led 0-to-1 production delivery of an enterprise architecture solution blueprint platform built with Node.js, TypeScript, and React; delivered a rules-based recommendation engine, auto-certification workflow, and real-time data rendering that maps 1,000+ architecture patterns to compliant component selections.
 - Agentic Architecture POCs:** Building ASB Assist, a natural-language architecture generation layer on top of ASB Builder, with MCP gateway and agent-gateway patterns that auto-generate compliant diagrams and service recommendations from chat prompts while injecting OAuth, Zero-Trust, and security controls.
 - AI Governance & Security Architecture:** Defined and operationalised enterprise AI governance covering Responsible AI, IAM, audit trails, regulatory compliance, and sovereign-cloud constraints across AWS, GCP, and OCP; the framework was adopted org-wide.
 - Observability & Developer Enablement:** Standardised OpenTelemetry log schemas and onboarding guides used by 20+ engineering teams, enabling vendor-agnostic telemetry, reusable integration patterns, and cleaner sample implementations across the platform.
 - Samples & Reference Implementations:** Created reusable internal examples and implementation guidance for secure service integration patterns so teams could adopt new platform capabilities faster with fewer one-off decisions.
 - Team Leadership:** Led 5 engineers with weekly sprint cadence, hands-on code reviews, ADRs, mentorship, and cross-functional alignment with product, risk, and architecture stakeholders.
- HealthCard** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - Identity Verification Platform:** Joined as first engineer on a pre-insurance payment platform and built the full product from scratch: React/TypeScript frontend, Node.js/Express REST API, Hyperledger Fabric verification layer, and AWS EKS deployment with HIPAA-compliant audit logging.
 - Fraud-Resistant Validation:** Designed the insurance validation and trust-score algorithm that supported real-time verification and helped the platform scale to a 20+ engineer team before acquisition.
 - Full-Stack Delivery Discipline:** Owned architecture decisions across frontend, API, and deployment layers, balancing speed, maintainability, and security controls in a small-team 0-to-1 environment.
- Dalhousie University** Halifax, Canada
Teaching Assistant & Research Assistant Jan 2021 – Aug 2022
 - Teaching Support:** Supported Software Development, Algorithms, and Advanced Cloud Computing courses by leading lab sessions, clarifying assignments, and reinforcing strong programming and systems fundamentals.
 - Research Support:** Worked on call-stack decision algorithms for compiler and runtime research, strengthening my systems thinking and ability to reason about lower-level execution details.

PROJECTS

- Agent Authorization Gateway:** Built a prototype OAuth/OIDC policy layer for MCP and A2A agent access, with audit logging, reusable SDK-style integrations, and phishing-resistant authorization flows for developer demos.
- DCS-BBN: Blockchain-Based Cloud Storage:** Designed a decentralized storage network using Solidity, Hyperledger Fabric, and zk-SNARK proofs for privacy-preserving verification; published in Springer ICMR.
- Enterprise Auth Sample Pack:** Assembled a set of secure identity integration samples showing login, token handling, and service-to-service authorization patterns for internal developers and architecture reviews.

EDUCATION

- University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- Dalhousie University** Halifax, Canada
Master of Applied Computer Science; GPA: 3.9/4.0 Jan 2021 – Aug 2022
- Gujarat Technological University** India
BE Computer Engineering; GPA: 3.92/4.0 Aug 2016 – Aug 2020